

Urban Rooftop Gardening Co-op App

Lab 3 — Architecture Selection & Justification

Architecture Selection

We chose **Layered Architecture** for the Urban Rooftop Gardening Co-op App.

Architectural Choice

The system is organized into three logical layers: a Presentation Layer for user interaction, a Business Layer for rooftop plot, crop and harvest, surplus produce, and community/soil-tip operations, and a Data Layer for persistent gardening information. This follows the layered structure specified in the Lab 3 handout.

Reason 1 — Separation of Responsibilities

The application contains several distinct responsibilities, including registering and managing rooftop plots, logging seasonal crops, calculating harvest windows and expected yield, managing surplus produce, and sharing soil-maintenance information. Separating these responsibilities into layers makes the architecture easier to understand and maintain.

Reason 2 — Easier Maintenance and Modification

Changes to the user interface can be made without directly changing database operations. Similarly, business rules such as harvest calculations or surplus-listing logic can be modified independently of the presentation layer. This supports the modular organization required by the system.

Security Advantage

The Presentation Layer does not directly access the database. Requests are handled through the Business Layer, which provides a controlled point for validating user actions and enforcing access rules before data is read or modified.

Performance Benefit

The separation of presentation, business processing, and data access allows frequently used operations to be optimized within their respective layers. Database queries can also be managed independently from user-interface processing.

Components in the UML Diagram

Layer	Component
Presentation	User Interface
Business	Plot Management
Business	Crop & Harvest Management
Business	Surplus Produce Management
Business	Community & Soil Tips
Data	Rooftop Gardening Database

Interfaces shown: Plot API, Crop/Harvest API, Surplus API, Community API, and database data interfaces for the corresponding business components.